Behavioural Cloning, DAgger and GAIL for RoboCup Walking: Comparisons Across Training Resource Allocations

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***Abstract*—Behavioural cloning (BC), dataset aggregation (DAgger) and generative adversarial imitation learning (GAIL) differ in their use of expert demonstrations, expert queries, environment interaction and optimisation. We compare them on simulated humanoid walking in the RoboCup 3D league, imitating the FC Portugal walking controller through its native gait interface. Across 3,205 unique 60-second deployments, paired on shared evaluation rounds, we measure task progress, imitation fidelity and stability on the original field and on fields with altered surface slip. Every supervised-initialised learner's mean progress is within 1.8% of the expert's, while untrained networks and zero actions driving the same gait engine make no forward progress. On the original-field training lineage, the fidelity ranking depends on which resource is held fixed: DAgger has 7.2× lower held-out action error at the initial fixed budgets, BC is 33.3% lower at matched optimiser updates, and DAgger is 17.7% lower under a common validation-stopping rule while using 66.3% more updates. After changes to its reward and update rules, GAIL refinement changes its initialiser's held-out action error by at most 2.4%. These results concern one task, one gait interface and the tested allocations. We recommend that comparisons of imitation methods state which resource they standardise and report results under more than one standardisation.**

***Index Terms*—imitation learning, behavioural cloning, DAgger, GAIL, humanoid locomotion, RoboCup, experimental design**

I. Introduction

RoboCup teams develop low-level skills such as walking and kicking by hand engineering and by optimisation, including reinforcement learning (Ingram et al., 2023; Beukman et al., 2024; Abreu et al., 2025). Imitation learning offers a way to train a policy from such an existing controller. A learned copy can initialise reinforcement learning, be distilled into a different architecture or be extended with new inputs, such as the surface information we test here. We consider three methods. Behavioural cloning (BC) fits expert actions by supervised learning. DAgger queries the expert at states the learner itself visits, to correct the compounding errors that follow when a cloned policy drifts away from the demonstrations (Ross et al., 2011). GAIL matches the expert's state-action distribution through an adversarially learned reward and reinforcement learning (Ho and Ermon, 2016). Ho and Ermon (2016) motivate GAIL by noting that BC "only tends to succeed with large amounts of data, due to compounding error caused by covariate shift".

Spencer et al. (2021) report that how much interaction helps depends on the setting. A comparison is also shaped by how resources are allocated, because the methods draw on overlapping but unequal resources. All three use expert demonstrations and optimiser updates. DAgger additionally requires environment interaction to visit learner states and expert queries to label them, and refits a growing dataset. GAIL additionally requires environment interaction for policy optimisation and the training of a discriminator and critic, and in our setting inherits the costs of its initialiser. Our comparisons control selected resources, they do not equalise all costs.

We study this in the RoboCup 3D simulation league, imitating the walking controller of the FC Portugal team (Abreu et al., 2025) on a simulated NAO humanoid. We ask three questions:

- **RQ1 (stability):** do BC, DAgger and GAIL differ in fall rate and kinematic smoothness while walking usefully?
- **RQ2 (efficiency):** how do imitation fidelity and task performance vary across the tested allocations of expert labels, environment interaction and optimisation?
- **RQ3 (expert comparison):** how close are the learners to the expert in progress and velocity, and does closer imitation improve these outcomes?

We additionally test the learners on fields with altered surface slip, and whether providing the slip value as an input changes the measured outcomes.

Our contributions are: (i) a comparison of the three methods across 3,205 unique deployments, paired on shared rounds, with controls testing whether zero actions or untrained networks walk through the gait engine, (ii) measurements showing that the BC and DAgger fidelity ranking reverses twice across three standardisations of resources on one original-field training lineage while mean progress stays close to the expert's, (iii) a record of the changes made to GAIL after a first implementation diverged, after which GAIL changes its initialiser's held-out action error by at most 2.4%, and (iv) recommendations for reporting resource allocations.

II. Background

A. Distribution shift and when interaction helps

Pomerleau (1989) observed that a cloned driving network "must not solely be shown examples of accurate driving, but

also how to recover". Ross et al. (2011) formalised the problem: a policy trained only on expert states can encounter states the expert never visited, and its errors can compound over the horizon. DAgger aggregates expert labels at learner-visited states. Its analysis relates policy performance to supervised loss under the assumptions of the no-regret reduction (Ross et al., 2011). Supervised fitting error is therefore relevant when comparing trained policies. Additionally, the cost of a mistake depends on how quickly the system can recover: if the expert returns to its normal state distribution within a few steps, the relevant constant "will be $O(1)$" rather than growing with the horizon (Ross et al., 2011, §2.2).

Spencer et al. (2021) distinguish regimes of covariate shift. In the simplest, the expert is realisable by the learner's policy class, and they report that commonly used benchmarks "are well solved by naive behavioral cloning" in this regime. Celemin et al. (2022) evaluate the data efficiency of interactive methods as "the amount of human data (i.e., feedback) that is required".

B. Adversarial imitation

GAIL trains a discriminator to separate expert and policy state-action pairs and uses its output as a reward for policy optimisation (Ho and Ermon, 2016). Its original evaluation used trust-region updates and suggested "initializing policy parameters with behavioral cloning" to reduce environment interaction. For generative adversarial networks, Goodfellow et al. (2014) note that the minimax objective saturates early in training, when the discriminator rejects generated samples with high confidence, and propose a non-saturating alternative. Orsini et al. (2021) show that adversarial imitation is highly sensitive to such implementation choices, including input normalisation, reward form and the handling of episode termination, and that BC pretraining gives little benefit in most of their environments.

C. Resource allocation in published comparisons

Published comparisons fix resources in different ways. Ho and Ermon (2016) gave the adversarial and apprenticeship methods "exactly the same amount of environment interaction" but trained their BC baseline "until validation error stops decreasing". Ross et al. (2011) plot performance against the number of training labels collected. Codevilla et al. (2019) report that behaviour-cloning results are sensitive to initialisation and data order, and report means over repeated runs because their simulator is non-deterministic. Hussenot et al. (2021) show that the choice of validation criterion itself affects which imitation policy is selected.

D. RoboCup 3D

The RoboCup 3D league simulates NAO humanoids in the rcssserver3d simulator. Teams build low-level skills by hand engineering and optimisation, increasingly with reinforcement learning, RobocupGym exposes such tasks with baselines trained for millions of environment steps (Beukman et al., 2024). FC Portugal's skills are learned policies that act through skill-specific primitives rather than directly on the joints (Abreu et al., 2025), its walk outputs gait parameters to a gait generator and inverse kinematics. This controller is our chosen expert. It supplies unlimited demonstrations, a condition under which Ho and Ermon (2016) report satisfactory behavioural cloning, and it can be queried at any state at no cost, which removes DAgger's usual labelling cost. Lastly, Ingram et al. (2023) report that a kicking agent penalised for falling learned to stand completely still to avoid the penalty. A controller can therefore record few falls without attempting the task, so we report falls together with task progress.

III. Methodology

A. Task, expert and policy interface

The expert is the unchanged FC Portugal walking controller (Abreu et al., 2025). At each 20 ms step it maps 63 observation features (body state and target information) to 16 native gait actions. A stateful gait generator and inverse kinematics convert those actions into joint commands for the robot's 22 joints. Every learner uses the same interface: a $63 \to 256 \to 256 \to 16$ ReLU network producing the 16 gait actions. The expert network is smaller than the learners (one 64-unit tanh hidden layer). The gait generator and inverse kinematics are identical for the expert and all learners.

Each episode walks towards a sequence of targets inside one half of the field. A fall is detected when head height drops below 0.28 m or the torso tilts beyond 60° in roll or pitch, and triggers a common reset and stabilisation. Reset time is excluded from the 3,000 active steps (60 s) of each deployment, so fall counts measure detected balance losses rather than autonomous recovery.

B. Learners

BC trains on 170,000 expert state-action pairs (balanced long-walk and short-turn episodes), with separate 10,000-example validation and test sets, using Adam (learning rate 10^{-4}, batch 256) and mean squared error over the 16 actions.

DAgger starts from a policy fitted on 100,000 balanced expert examples, then performs seven rounds. Each round collects 10,000 steps in which the expert controls the robot with probability β (β = 0.8, 0.6, 0.4, 0.2, 0, 0, 0 over the seven rounds) and otherwise the learner does, the expert labels every visited state, the policy continues training from its current weights on the full aggregate for 50 epochs. The final aggregate has 170,000 labels, the same number as BC's training set, but 70,000 are labels collected during the expert-learner mixture rollouts (learner-only in the final three rounds).

GAIL uses the 170,000 expert examples as the expert distribution and a PPO actor-critic (Schulman et al., 2017) trained on 70,000 learner environment steps. We report two supervised-initialised variants, GAIL-BC and GAIL-DM (initialised from BC and from DAgger respectively), and a random-initialised control, run on the original field and for the three varied-only hidden seeds. A first implementation with the minimax reward $-\log(1 - D)$ diverged (Section IV-D). The final implementation uses the non-saturating reward $r(s, a) = \log D(s, a)$ (Goodfellow et al., 2014), discriminator

inputs normalised by expert statistics, 0.9/0.1 label smoothing, an interpolation gradient penalty, discriminator early stopping once both class accuracies exceed 0.95, a much smaller actor learning rate, target-KL early stopping with rollback of over-large updates, and a decaying auxiliary BC loss that anchors the actor to its initialiser. We therefore describe this method as BC-anchored GAIL refinement, our conclusions concern this implementation. Its settings are $\gamma = 0.99$, $\lambda = 0.95$, PPO clip 0.2, target KL 0.02, learning rates of 3×10^{-6} (actor), 3×10^{-4} (critic) and 10^{-4} (discriminator), gradient-penalty weight 10, initial policy standard deviation 0.14 and a BC-anchor weight decaying from 0.05 to 0.02 over the seven iterations. A fall ends the episode. Because $\log D < 0$, ending an episode early removes future negative reward, a known bias of this reward under termination (Kostrikov et al., 2019).

C. Surfaces and training stages

The original field is kept as installed. Altered surfaces set a single contact-slip scalar for the whole field. Besides the original-field models, we train two model families on varied surfaces, cycling through slip values 0.025, 0.1, 0.2, 1 and 10. *Continued* models start from the corresponding original-field model and continue training on varied surfaces, *varied-only* models are trained on varied surfaces from scratch. Each family has two variants: *hidden* models receive no surface information, and *input* models receive the normalised slip value as one extra input (64 inputs). Varied-surface models are trained with three seeds (42,43 and 44), original-field models have a single training lineage. Every family uses the original-field budgets (Section III-B): 170,000 expert examples, a 100,000-example DAgger seed, seven 10,000-step learner rounds and 70,000 GAIL steps, with the expert data and every learner batch balanced across the five slip values. GAIL-BC and GAIL-DM start from the same family's BC and DAgger models. Varied-only BC is also trained for 200 epochs in total, and this model is its initial model in Figure 2.

Evaluation on altered surfaces uses a fixed suite of 30 rounds (15 exploratory and 15 confirmation, pooled in all analyses). Each round draws one slip value from a truncated Gaussian centred on one of the five training values (standard deviation half the mean, support $(0, 10]$, realised values 0.0003 to 8.96) and one target sequence. Every controller is evaluated on the same 30 rounds, which makes comparisons paired. The original field is evaluated on five target seeds.

D. Resource accounting and the three standardisations

Table I lists the recorded training resources for each learner. We compare BC and DAgger under three standardisations. (1) *Initial fixed budgets*: BC for 100 epochs, DAgger for its seven 50-epoch rounds, these equalise data counts but not optimisation. (2) *Matched optimiser updates*: BC is retrained from the same initialisation for exactly DAgger's 211,050 updates, a count that includes DAgger's 50-epoch seed fit. (3) *Common stopping rule*: both methods resume from their checkpoints on their fixed final data and stop when validation error has not improved by at least 10^{-5} for 25 epochs (cap 1,000 epochs, never reached), with validation-stopped BC resuming from the matched-updates checkpoint. We call these models *validation-stopped*, this is not a proof of convergence, and for DAgger it is further fitting of the saved aggregate without new expert queries. The same rule is applied to 24 varied-surface models (BC and DAgger, continued and varied-only, hidden and input, three seeds), giving 26 validation-stopped models in total, this further fitting adds 6,476 epochs and 4.3 million updates.

E. Measurements

Task. Progress is the distance covered towards successive targets per 60 active seconds. For the 26 validation-stopped models on the 30 altered-surface rounds, shared rounds explain 99.74% of the progress sum of squares. We therefore compare controllers through paired differences on the same round.

Fidelity. Held-out MSE compares the learner's 16 actions with the expert's on the expert test set. On-policy MSE compares them on states the learner visits during deployment.

Stability. Falls per active minute, and a jerk rate: the squared third derivative of all 22 joint angles (after a five-sample moving average), summed over joints and averaged over active time, in rad^2/s^6. Simulator torques are not logged, so jerk is kinematic, not a measure of energy or wear. The tilt condition in the fall detector was added after early GAIL runs left the robot lying with its head above the height threshold (roll RMS 92°, zero falls recorded).

Controls. A zero-action controller, three untrained networks with the learners' architecture, and uniform random actions all drive the same gait engine, testing whether those specific outputs produce useful walking through the shared engine.

Some deployments belong to more than one evaluation cohort, the 3,205 deployments count each unique deployment once, and the cohorts are not independent replications of a single comparison. Original-field summaries use five target seeds, altered-surface contrasts use 30 shared rounds, with three training seeds per varied family. Unless stated otherwise, intervals are 95% paired bootstrap intervals over rounds (5,000 resamples, training seeds averaged within each round). They condition on the trained checkpoints and do not describe a population of training runs. Closeness to the expert is reported descriptively, no equivalence margin or formal equivalence test is specified. Percentages and ratios are computed from unrounded values.

IV. Results

We report RQ2 first, because its standardisations are used in RQ3 and RQ1.

A. RQ2: the fidelity ranking depends on the standardisation

Figure 1 and Table II show the central result: which method has the lower held-out error depends on how the training budget is set (Section III-D). (1) At the initial fixed budgets, DAgger's held-out error is 7.2× lower than BC's (0.0153 against 0.1105). These budgets give DAgger 211,050 updates and BC 66,500, at similar update counts the two learning curves differ by

TABLE I

RESOURCES CONSUMED BY EACH LEARNER ON THE ORIGINAL FIELD. MATCHING UPDATES LEAVES QUERIES AND INTERACTION UNEQUAL, A COMMON STOPPING RULE FIXES THE RULE BUT NOT THE COST. UPDATES ARE TOTAL PERFORMED LINEAGE COSTS, INCLUDING FITTING AFTER THE SELECTED CHECKPOINT. EXPOSURES COUNT USES, NOT UNIQUE LABELS. GAIL REUSES THE EXPERT DATASET AND INHERITS ITS INITIALISER'S COSTS, REUSE DOES NOT ADD UNIQUE DEMONSTRATIONS.

Learner	Static expert examples	Expert labels in mixed rollouts	Collection / learner env. steps	Optimiser updates	Exposures	Stopping rule
BC, initial	170k	0	0	66,500	17.0M	100 epochs
BC, matched updates	170k	0	0	211,050	54.0M	fixed updates
BC, validation-stopped	170k	0	0	288,855	≈74M	val. patience
DAgger, 7 rounds	100k	70k	70k	211,050	≈54M	50 ep./round
DAgger, validation-stopped	100k	70k (no new)	70k (no new)	480,375	≈123M	val. patience
GAIL-BC / GAIL-DM	170k reused	0 / 70k inherited	70k / 140k total	inherited + PPO	—	70k env. steps

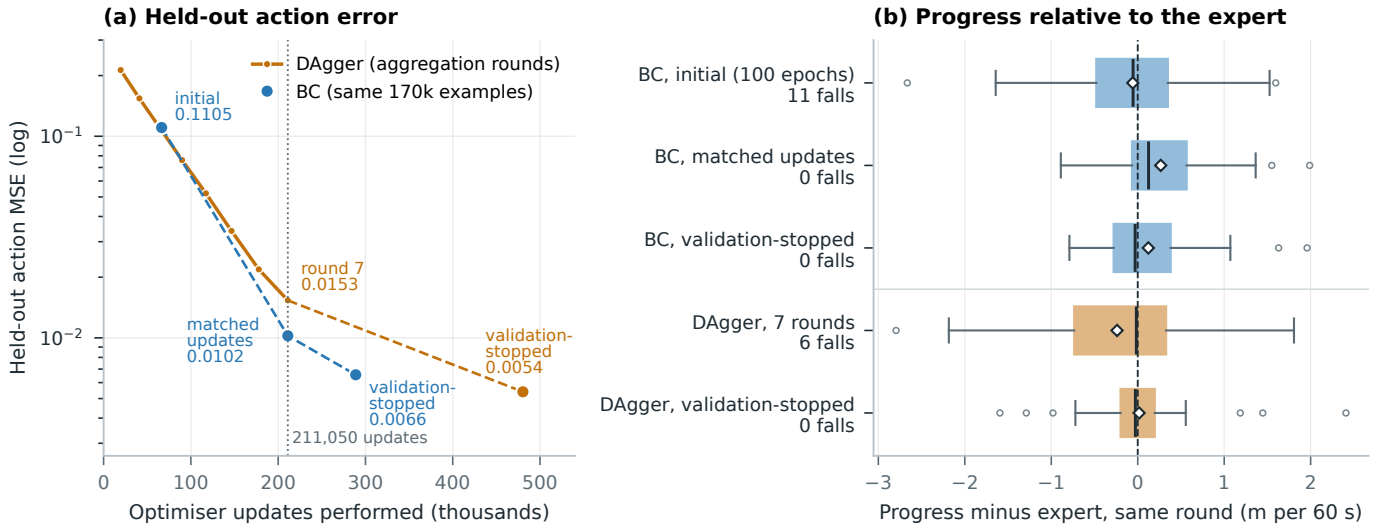


Fig. 1. Original-field models, one training lineage. (a) Held-out action error against optimiser updates performed. DAgger points are its saved aggregation rounds, dashed segments are further fitting on fixed final data. (b) Paired progress difference from the expert on the 30 altered-surface rounds, box = interquartile range, line = median, ◇ = mean, whiskers extend to the most extreme observations within 1.5×IQR of the quartiles, circles = rounds beyond the whiskers. Falls are totals over 30 active minutes.

only 0.5%. (2) With matched updates the ranking reverses: BC's held-out error is 33.3% lower than DAgger's (0.0102 against 0.0153), and its on-policy error is also lower (0.0040 against 0.0078). (3) Under the common validation-stopping rule it reverses again: DAgger's held-out error is 17.7% lower (0.0054 against 0.0066) and its on-policy error 8.2% lower, but DAgger needs 66.3% more updates (480,375 against 288,855).

None of the three comparisons therefore identifies a method that is better in general. Task progress on the altered-surface suite hardly differs across them (Figure 1(b)): mean paired differences from the expert lie between -0.24 m (seven-round DAgger) and $+0.27$ m (matched BC), that is, within -0.7% to $+0.8\%$ of the expert's progress.

In the varied-surface families, BC's gap closes with further fitting (Figure 2). After initial training, BC's held-out error is 4–11× DAgger's across the four paired comparisons. After validation-stopped fitting the ratio is 0.9–1.1×, and for the continued input models BC's error is the lower of the two.

TABLE II

ORIGINAL FIELD, MEAN OF FIVE DEPLOYMENTS. MSE IS OVER THE 16 GAIT ACTIONS, PROGRESS IN METRES PER 60 S.

Controller	Updates	Held-out MSE	On-policy MSE	Progress
Expert	—	—	0	44.45
BC, initial	66,500	0.1105	0.0744	44.63
BC, matched	211,050	0.0102	0.0040	43.91
BC, val.-stopped	288,855	0.0066	0.0021	44.17
DAgger, 7 rounds	211,050	0.0153	0.0078	44.36
DAgger, val.-stopped	480,375	0.0054	0.0019	43.98
GAIL-BC	BC + PPO	0.1104	—	44.56
GAIL-DM	DAgger + PPO	0.0157	—	44.15

BC needs more fitting to get there: continued BC stops after 217–448 further epochs, continued DAgger after 25–70.

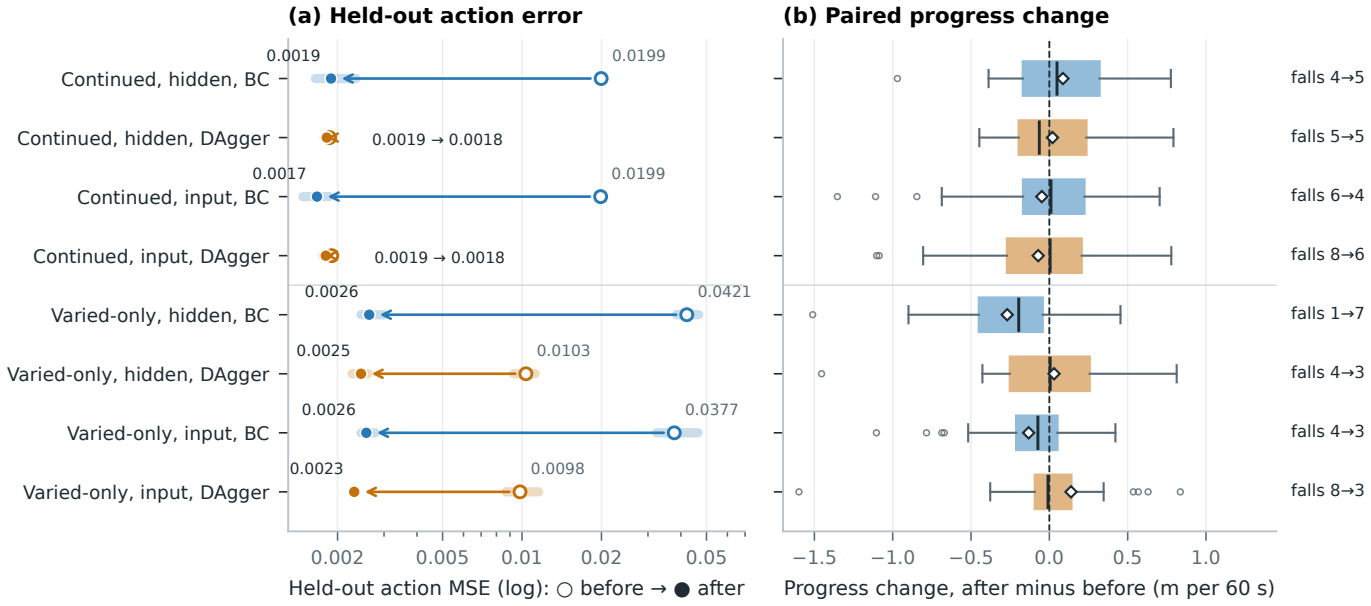


Fig. 2. Varied-surface families, three training seeds each. (a) Held-out action error before (○) and after (●) validation-stopped fitting, averaged over seeds, shaded bars show the range across the three seeds. (b) Distribution over the 30 altered-surface rounds of the paired progress change (after minus before, averaged over seeds within each round), box = interquartile range, line = median, ◇ = mean, whiskers extend to the most extreme observations within 1.5×IQR of the quartiles. These are distributions of round-level changes, not confidence intervals, intervals for the mean are reported in the text. Fall counts are totals over 90 active minutes (three seeds × 30 runs) before → after.

GAIL refinement barely changes its initialiser. Held-out error changes by -0.05% for GAIL-BC and $+2.4\%$ for GAIL-DM (Table II), median paired progress differences from the initialiser stay within ± 0.11 m, and all eight varied-family intervals include zero. From a random start, 70,000 steps produce no walking: no targets are reached, and progress on the original field is -0.12 m. During training, the four random-initialised actors fall 65 times in 280,000 steps, against 2 and 6 falls in 910,000 steps for the 13 GAIL-BC and 13 GAIL-DM runs. A random actor falls often regardless, so these counts do not show whether the termination bias of the $\log D$ reward (Section III-B) contributes.

DAgger's seed dataset also mattered: replacing an earlier 20,000-example seed with 100,000 balanced examples reduced held-out error by 78.5% and on-policy error by 95.0%, although this changed both the size and the composition of the seed.

B. RQ3: comparison with the expert

All supervised-initialised learners walk about as well as the expert. On the original field, their mean progress is within -1.2% to $+0.4\%$ of the expert's 44.45 m (Table II), and their forward velocity within -1.6% to $+0.4\%$ of its 0.759 m/s. On the altered-surface suite, all supervised-initialised model families (averaged over training seeds) are within -1.2% to $+1.3\%$ of the expert's 32.89 m, and individual seed models within -1.2% to $+1.8\%$.

The controls show that this walking comes from the learned actions, not from the gait engine alone (Figure 3). On the altered-surface suite, zero actions and the three untrained networks make no progress (-0.64 to -0.16 m), and uniform random actions fall 704 times in 30 minutes. Fall counts alone would not reveal this: on the original field, the zero-action controller and the untrained networks record no falls.

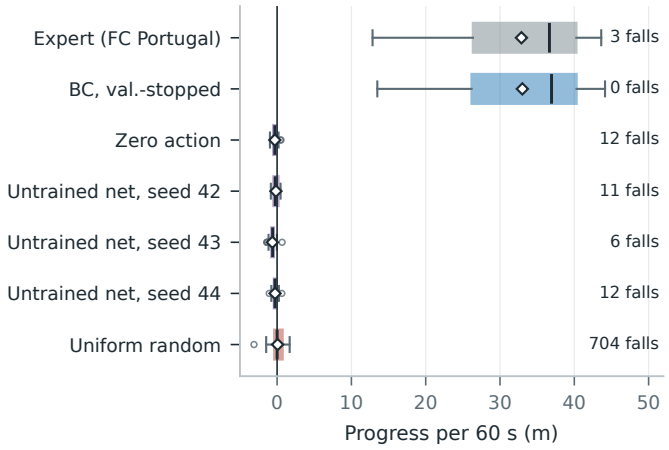


Fig. 3. Progress per 60 s on the 30 altered-surface rounds for the expert, a trained policy and controls that drive the same gait generator, inverse kinematics and resets, box conventions as in Fig. 1. Right: total falls in 30 active minutes. The controls move much less than walking controllers, so their fall counts should be read alongside progress, Fig. 4 counts runs with at least one fall.

Lower action error does not bring higher progress. Validation stopping cuts varied-only hidden BC's held-out error by 93.7%, yet its progress drops by 0.27 m per 60 s $[-0.38, -0.17]$ and its falls rise from 1 to 7 over 90 minutes, varied-only input BC loses 0.13 m $[-0.23, -0.03]$. Both losses are under 1% of progress, but their conditional intervals exclude zero. Likewise, validation-stopped DAgger has lower action error than validation-stopped BC, yet makes 0.11 m less progress on

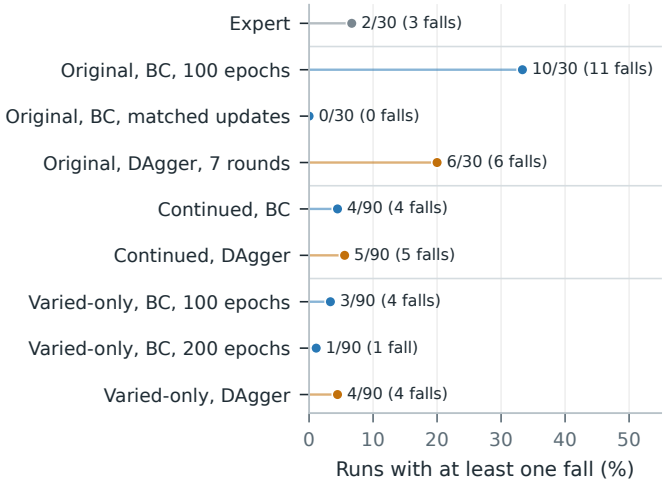


Fig. 4. Share of runs with at least one fall on the 30 altered-surface rounds, for fixed-budget hidden models, labelled with runs and total falls. Original: trained on the original field only (30 runs), continued and varied-only families have three training seeds (90 runs). No intervals are shown because runs share rounds and training histories, so they are not independent.

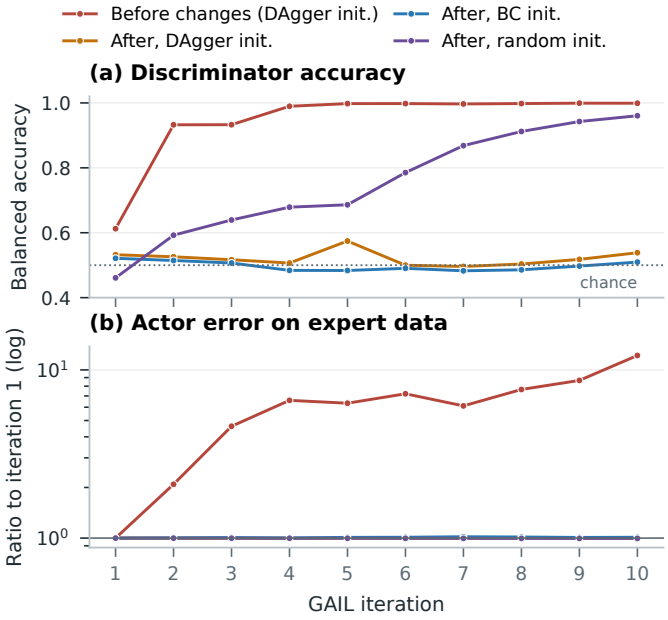


Fig. 5. Diagnostics from the historical GAIL development runs before and after the changes to its reward and update rules, the DAgger initialiser is the earlier DAgger lineage. (a) Discriminator balanced accuracy (dotted line: chance). (b) Actor error on expert training data as a ratio to its iteration-1 value, after the changes, all three lines overlap near 1. Rollouts are 4,000 steps, these are diagnostics, not checkpoint rankings, and all changes were introduced together.

the altered-surface suite $[-0.26, +0.04]$.

C. RQ1: stability

No method shows a consistent stability advantage (Figure 4). Pooled over training histories (hidden context, fixed budgets with 100-epoch BC, 210 runs each), runs with a fall number 17 for BC, 15 for DAgger, 14 for GAIL-BC and 11 for GAIL-DM, against 2 of 30 for the expert, and the median jerk rate is 40.1–41.7 million rad^2/s^6 for all four methods (expert 41.6), with overlapping interquartile ranges. The clearest difference is within one BC lineage: initial 100-epoch BC falls in 10 of 30 altered-surface runs (11 falls), and matched-update BC, with the same data, architecture and initialisation, in none. After validation stopping, the two original-field models record no falls, and the varied families 3–7 falls per 90 minutes. During training, DAgger data collection never falls in 910,000 steps, including the rounds with $\beta = 0$.

Jerk mostly tracks the shared round conditions. Across the 26 validation-stopped models and 30 rounds, progress and jerk are strongly correlated (Spearman $\rho = 0.93$), but only weakly ($\rho = 0.32$) once each round's mean is removed. Within rounds, action error and jerk are negatively correlated ($\rho = -0.77$).

D. GAIL before and after the training changes

The runs in this subsection are historical development runs. Their actor error is mean squared error on the expert training data, a training diagnostic that is not comparable with the held-out MSE in Table II.

Before the changes, the first implementation (DAgger-initialised, minimax reward, no KL control) diverged within ten iterations (Figure 5). The discriminator became almost perfect (balanced accuracy 0.61 to 0.999), the learned reward collapsed (0.65 to 0.0017), updates were very large (approximate KL 215–1,203, with 98.6–98.7% of PPO samples clipped), actor error on expert data grew 12.2×, and the robot toppled.

After the changes (Section III-B), training is stable: balanced accuracy stays at 0.48–0.57 and accepted KL at 0.0077–0.0116, without rollbacks. The supervised-initialised actors barely move: error on expert data stays within 0.2678–0.2683 for the DAgger-initialised run (an earlier DAgger lineage, not the policy that initialises GAIL-DM) and within 0.00705–0.00719 for the BC-initialised run. The random-initialised run does not diverge but reaches no targets.

All changes were introduced together, so neither the absence of divergence nor the small movement of the supervised-initialised actors can be attributed to any single change, several of them (the BC loss, the smaller actor learning rate and the KL limit) constrain how far the actor moves by design.

E. Altered surfaces and explicit context

Slip reduces everyone's progress but not the learners' gap to the expert. The expert's median progress falls from 42.6 m in the group centred on 0.025 to 18.1 m in the group centred on 10, while the learners' paired gap stays centred near zero in every group (Figure 6). Slip shows up instead in falls (43 of 354 runs with a fall in the group centred on 10, against 3–20 of 354 in each of the other groups) and in BC's and DAgger's on-policy action error, which is highest in the group centred on 10.

Giving the policy the slip value does not consistently help. After validation stopping, the input-minus-hidden progress interval includes zero for three of four families. The exception, varied-only BC ($+0.19$ m $[+0.07, +0.32]$), coincides with the

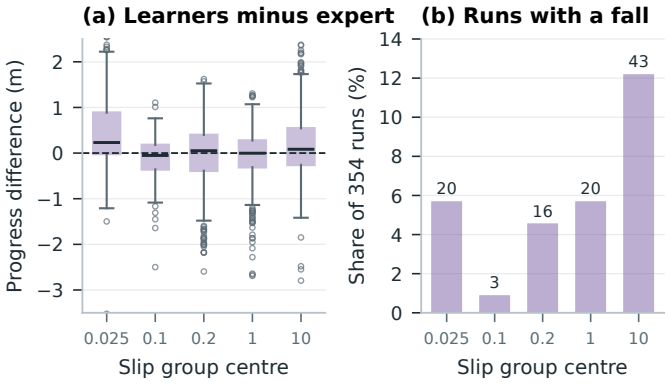


Fig. 6. Altered surfaces in the earlier evaluation cohort (all trained BC, DAgger, GAIL-BC and GAIL-DM controllers, all training histories and both context types, validation-stopped models not included), grouped by the slip value around which each round's slip is drawn. Each group has six rounds and 354 runs, the group centred on 10 contains sampled slips, not a fixed slip of 10. (a) Paired progress difference from the expert per 60 s, box conventions as in Fig. 1. (b) Share of runs with at least one fall, labelled with counts. Rounds vary the target sequence as well as slip.

hidden model's loss after further fitting and is not corrected for multiple comparisons. In an earlier BC study (90 paired runs), supplying a wrong slip value barely changes behaviour: median on-policy error $+1.3\%$, mean progress $+0.01$ m, and 4 falls against 4. Beukman et al. (2023) note that a context-unaware policy can perform well when the same policy makes progress in every context, here the expert has no slip input and makes progress on every tested surface.

F. Representations

In an exploratory extension related to RQ3, we ask whether hidden representations explain why lower action error does not bring more progress. Probe scores do not track deployment performance. We fit linear ridge probes (Alain and Bengio, 2016) on a shared bank of expert trajectories, covering 131 representations and 30 targets, with raw-input, random-network and three shuffled-label controls (Hewitt and Liang, 2019) and CKA comparisons (Kornblith et al., 2019). Targets describing future outcomes are computed under expert actions, not each learner's deployment. For example, the untrained networks, which cannot walk, decode command displacement with $R^2 = 0.97$, and DAgger's phase-roll probe score decreases (0.712 to 0.694) while its action error falls by 65%, with its hidden layers keeping a CKA of 0.99 with its initialiser. We use these results only descriptively.

V. Discussion

A. The choice of standardised resource

The tested BC, DAgger and GAIL configurations have different resource requirements, and our allocations do not equalise all of them (Table I). On the original-field lineage, the fidelity ranking reverses twice across the three standardisations, while mean paired progress stays within -0.7% to $+0.8\%$ of the expert's under all three. Prior comparisons also fix different resources for different methods: Ho and Ermon (2016) match environment interaction among adversarial methods but validation-stop their BC baseline, and Ross et al. (2011) compare methods per collected label. Inherited costs are a further unequal quantity: GAIL inherits its initialiser's demonstrations, updates and (for GAIL-DM) expert queries, and continued models inherit their original-field training.

We suggest that comparisons of imitation methods state which resource is held fixed, account for inherited costs, and show results under more than one standardisation, or as outcome-against-cost curves, since here the choice of standardisation changes the fidelity ranking.

B. Relation to conditions identified in prior work

Prior work suggests possible explanations for similar progress across methods, but our experiments do not identify a mechanism. Spencer et al. (2021) discuss realisability of the expert within the learner's policy class. Our expert is a smaller network than the learners, but this alone does not establish realisability. Ho and Ermon (2016) report satisfactory BC performance with sufficient data on several locomotion tasks. We do not vary BC's 170,000-example training set, so we cannot identify a data-sufficiency threshold. Ross et al. (2011) relate their performance bounds to the cost of errors and recovery conditions. We do not measure recovery to the expert's state distribution. The observed absence of falls during DAgger collection and the action-error ratios reported here do not establish that condition.

The largest differences in our data occur where one method has a smaller dataset or budget: DAgger with a 20,000-example seed, BC at 100 epochs, and GAIL from random weights with 70,000 steps. In contrast, Ho and Ermon (2016) report a large difference on their Humanoid task, where BC does not exceed 60% of expert performance and GAIL matches the expert. Their policies act directly on the simulator's joint actuators, whereas ours act through a gait generator, whether this difference in interface accounts for the different outcome is not tested.

Learning from few demonstrations, the setting in which Ho and Ermon (2016) report GAIL's advantage over BC, is not tested.

C. Limitations

The original-field comparison has a single training lineage, and continued families share original-field parents across seeds. Wall-clock time, expert compute and repeated training are not equalised. Our expert supplies programmatic labels without human noise, the setting Celemin et al. (2022) call "unrealistic perfect oracles". Rounds vary target sequences together with slip, so slip effects are not isolated. The wrong-context test covers BC only. Many contrasts are exploratory and uncorrected for multiplicity, and confirmation rounds had been examined before probing. All results are in simulation, for one skill and one robot model.

D. Future work

Three directions follow: varying GAIL's BC-loss weight, actor learning rate and interaction budget separately; settings

without the properties discussed above, such as fewer or imperfect demonstrations, an expert of greater capacity than the learner, external pushes, raw joint control or other skills; and wrong-context evaluation for all algorithms, with outcomes reported against each cost axis, including wall-clock time.

VI. Conclusion

On RoboCup 3D walking through FC Portugal's gait interface, behavioural cloning, DAgger and BC-anchored GAIL all reach the expert's level of task performance: every supervised-initialised learner's mean progress is within 1.8% of the expert's and its forward velocity within 1.6%, while zero actions and untrained networks driving the same gait engine make no progress. Closer imitation does not bring more progress, and no method shows a consistent stability advantage. What differs is imitation fidelity, and on the original-field lineage its ranking depends on which resource is held fixed: DAgger's held-out action error is 7.2× lower at the initial budgets, BC's is 33.3% lower at matched optimiser updates, and DAgger's is 17.7% lower under a common stopping rule at the cost of 66.3% more updates. GAIL refinement changes its initialiser's held-out action error by at most 2.4%. On altered surfaces, slip lowers the progress of the expert and the learners alike without widening the gap between them, and giving the policy the slip value does not consistently help.

These findings are restricted to the tested task, interface, datasets and training configurations. They show which outcomes differ across methods in this setting, but do not establish why progress remains similar. Here, how much optimisation a method receives matters more for imitation fidelity than which method is used: further fitting lowers BC's held-out error about 17-fold, whereas BC and DAgger differ by at most 1.5-fold at matched updates or under a common stopping rule, and task progress stays close to the expert's under every method and budget tested. For a RoboCup team with an existing walking controller, behavioural cloning trained for as many updates as DAgger therefore reproduces expert-level walking on the original field, with no falls on the altered-surface suite, without the expert queries or environment interaction that DAgger and GAIL add.

Data and reproducibility

All results come from saved checkpoints and evaluation records, no checkpoint is selected using test data. The figures and numbers in this paper are produced by scripts in the project repository from versioned summary files, and the repository includes a reproduction guide. Code is hosted on GitHub at https://github.com/TafaraC/Walker.

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